IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-145

1. Topic (12 words max)

Smart-Camp Box: An Integrated IoT and Artificial Intelligence Framework for Safety, Communication, Learning Assistance, and Resource Optimization in Disaster Relief Camps.

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-145
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Disaster relief camps are rapidly established in response to natural or human-made disasters; however, they frequently suffer from poor coordination and management in areas such as situational awareness, resource allocation, educational continuity, and risk prediction [1], [3]. The individuals and institutions most affected by these challenges include displaced populations, particularly children, elderly persons, and other vulnerable groups, as well as camp administrators, disaster relief organizations, and government authorities [10]. Current disaster camp management practices are often rudimentary or fragmented, lacking integration across systems. This results in overcrowding, shortages of essential resources, delayed medical assistance, increased health risks, and unmanaged education, especially in environments with low or unreliable network connectivity [3], [5].

These challenges create multidimensional impacts. Socially, inadequate camp management leads to insecurity, educational disruption, and loss of dignity among displaced populations [10]. Technically, the absence of integrated IoT and AI-driven systems prevents real-time monitoring, predictive analysis, and effective decision support [1], [4]. Economically, inefficient planning and lack of predictive resource management cause wastage or shortages, increasing operational costs for humanitarian agencies and governments [3], [5]. From an ethical, legal, and environmental perspective, poor planning exacerbates overcrowding and contributes to environmental degradation in and around relief camps [7], [10].

This research aligns with the FoC Research Cluster on Smart Systems, Artificial Intelligence, and Internet of Things (IoT) and targets key industry verticals, including Disaster Management, Humanitarian Technology, Smart Infrastructure, Education and Awareness [2], [4]. Through the proposed Smart-Camp Box, an integrated IoT–AI framework combining human and resource demand forecasting, environmental risk analysis, health monitoring support, and low-connectivity adaptive education, the research addresses critical functional gaps in existing disaster relief camp systems [6], [8], [9].

Furthermore, this research contributes directly to achieving several United Nations Sustainable Development Goals (SDGs), including SDG 4 (Quality Education), SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action), by enabling resilient, inclusive, and technology-driven disaster response systems [10].

[1] L. Atzori, A. Iera, and G. Morabito, “The Internet of Things: A survey,” *Computer Networks*, vol. 54, no. 15, pp. 2787–2805, 2010.

[2] A. Al-Fuqaha et al., “Internet of Things: A survey on enabling technologies, protocols, and applications,” *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347–2376, 2015.

[3] A. Rehman, M. H. Khan, and M. A. Shah, "IoT-based disaster management system," in *Proc. Int. Conf. Frontiers of Information Technology (FIT)*, 2017, pp. 271–276.

[4] S. Madakam, R. Ramaswamy, and S. Tripathi, "Internet of Things (IoT): A literature review," *Journal of Computer and Communications*, vol. 3, no. 5, pp. 164–173, 2015.

[5] S. Kumar, P. Tiwari, and M. Zymbler, "Internet of Things is a revolutionary approach for future technology enhancement: A review," *Journal of Big Data*, vol. 6, no. 1, 2019.

[6] Z. Ma, J. Yang, and W. Wang, "Crowd density estimation using deep learning," *IEEE Access*, vol. 6, pp. 24422–24432, 2018.

[7] A. Mehmood et al., "Machine learning-enabled IoT-based smart health monitoring system," *IEEE Access*, vol. 8, pp. 117094–117109, 2020.

[8] M. A. Khan, "A survey of Internet of Things-based smart healthcare," *IEEE Access*, vol. 8, pp. 113990–114006, 2020.

[9] S. Ghosh, A. Ghosh, and S. K. Das, "Adaptive learning systems: A review of techniques and applications," *IEEE Trans. Learning Technologies*, vol. 12, no. 2, pp. 208–221, 2019.

[10] United Nations High Commissioner for Refugees (UNHCR), *Emergency Handbook: Camp Coordination and Camp Management*, UNHCR, Geneva, Switzerland, 2022.

6. Brief description of Existing Research and Systems (300 words max)

Various existing products and research efforts have explored the use of Internet of Things (IoT), artificial intelligence (AI), and sensor technologies in the management of disasters. The requirements pertain to environmental monitoring, emergency notification, crowd analysis, and situation awareness, but not necessarily to camp management. For Instance, the smart city project, IoT-enabled Environmental Monitoring, and Public Safety Systems, in the likes of Barcelona and Tokyo, utilize IoT sensors and predictive analysis to detect flood-level data, seismic activity, and anomalous occurrences in the urban area for early notification and rapid response coordination. The data points towards real-time monitoring and predictive analysis, but its design and intent focus on the urban infrastructure, not the camp. The current product research in the area includes applications related to the utilization of the Internet of Things (IoT), Artificial Intelligence (AI), and Sensor Networks in the area related to the management of disasters. These systems demonstrate real-time monitoring and predictive capabilities but are designed for urban infrastructure rather than temporary camp environments.

In the realm of disaster management systems using the Internet of Things (IoT), the application potential of sensor technology has been found useful in real-time data acquisition and decision support systems related to disaster management. In fact, the applicability of IoT technology has been described regarding disaster management-related tasks such as data acquisition, situational awareness, and resource management, with improvements attributed to the use of various data from sensors being processed. Another application project on crowd monitoring and social distancing using the IoT has been demonstrated, related to crowd detection and risk levels based on simulated events.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

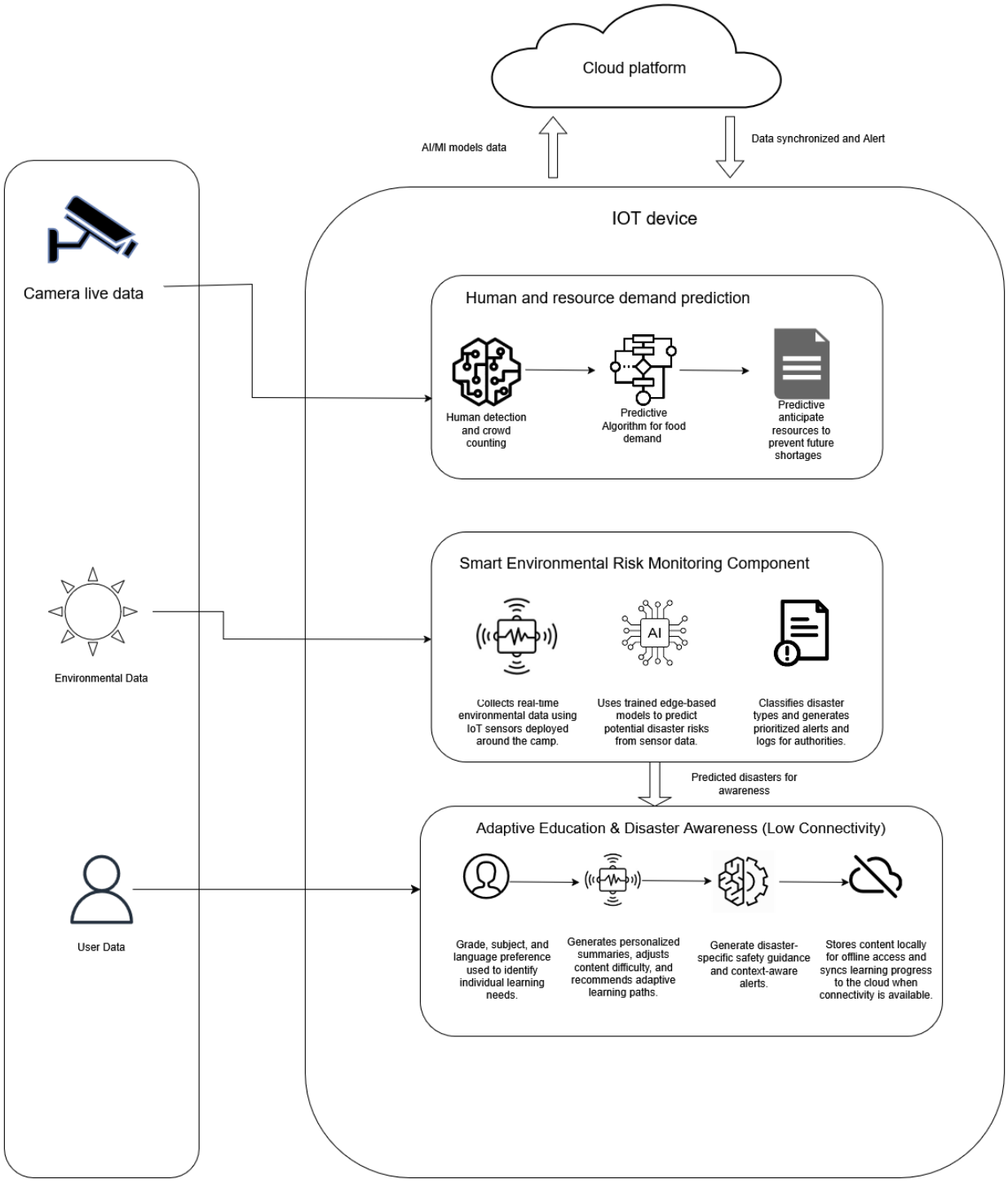
The proposed work seeks to develop a Smart IoT Box-based intelligent system, which would be beneficial to disaster relief camps, especially under conditions of limited or disrupted network connectivity. Resource management, disaster education, and environmental risk analysis would be addressed through IoT sensing, machine learning, and cloud analysis.

In the conceptual design diagram above, the role played by the Smart IoT Box is depicted as the main processing unit installed within the disaster relief camp infrastructure. It is responsible for processing data from a variety of sources, such as vision from cameras, environmental sensors, and data from users, even when there is no internet connection. This data is processed at the edge level even when there is no internet connection, but is synchronized once the internet connection is re-established.

The first significant part of the framework is the Human and Resource Demand Prediction module. This component uses camera-based IoT inputs for human detection and crowd counting through machine learning approaches. The information derived from the detected population is further processed to estimate their demographics, which are used to predict demands for necessities such as food, water, and medicine in real time and in the future. By making this prediction automatic, this system eliminates the need for estimation, which was being performed manually.

The “Smart Environmental Risk Monitoring” module is constantly gathering data from the environment sensors installed around the camps, such as temperature, rainfall, or water level sensors. Machine learning algorithms are implemented to identify irregularities to gauge disaster risks such as floods and landslides, among others. Once disaster risks are detected, the “Smart IoT Box” module is responsible for creating local notifications as well as sending vital disaster alerts to the cloud platform. In the Adaptive Education and Disaster Awareness module, there is an emphasis on developing awareness as well as preparedness among the displaced population members. Since the environment is low on connectivity, all the related content is either stored within the Smart IoT Box or accessed offline by the users

The content is dynamically generated depending on the context and risk predictions available within the disaster context, thus ensuring the relevant content is available at the appropriate time. Overall, the proposed solution brings a new, modular, intelligent, and more robust Smart IoT Box design, which helps disaster relief camps carry out their tasks more efficiently and safely. With the integration of edge intelligence analysis and cloud analytics, the proposed solution helps with data-driven decision-making, risk management, resource allocation, and awareness delivery even in a disaster scenario.


Cloud platform
AI/MI models data
Data synchronized and Alert
IOT device
Camera live data
Human and resource demand prediction
Human detection and crowd counting
Predictive Algorithm for food demand
Predictive anticipate resources to prevent future shortages
Environmental Data
Smart Environmental Risk Monitoring Component
AI
Collects real-time environmental data using IoT sensors deployed around the camp.
Uses trained edge-based models to predict potential disaster risks from sensor data.
Classifies disaster types and generates prioritized alerts and logs for authorities.
Predicted disasters for awareness
User Data
Adaptive Education & Disaster Awareness (Low Connectivity)
Grade, subject, and language preference used to identify individual learning needs.
Generates personalized summaries, adjusts content difficulty, and recommends adaptive learning paths.
Generate disaster-specific safety guidance and context-aware alerts.
Stores content locally for offline access and syncs learning progress to the cloud when connectivity is available.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The effective deployment of the proposed Smart IoT box for the disaster relief camp requires interdisciplinary expertise, knowledge, and data. The authors have collective expertise and academic knowledge related to the subject matter for handling the primary components for the proposed work, such as IoT technology, ML technology, cloud technology, as well as the application area pertaining to the management of the disaster relief camp.

In terms of technical expertise, the group possesses good foundational knowledge regarding the Internet of Things (IoT) technologies of sensor integration, programming of edge devices, as well as data acquisition. The previous experience of the group regarding the implementation of systems based on IoT technology is beneficial for the effective use of low-power sensors, cameras, as well as processing units that are required during a disaster scenario. Moreover, the group possesses experience with machine learning models like deep learning models of computer vision for human detection, time series models for forecasting, and classification models that are required for population estimation and resource allocation analysis.

The research also derives value from their expertise in cloud systems, storage, analytics, as well as alert systems on cloud platforms. They will be of great value in designing a hybrid edge cloud solution that will facilitate continuous operation of the system with reduced connectivity capabilities as well as long-term analysis upon connectivity access. Turning to the knowledge of the study area, the group of authors appears to possess good knowledge of the concepts of disaster management, relief camps, distribution of resources, importance of early warning systems, and awareness systems.

This fact is evidenced by the presence of existing literature on the issue of disaster management and relief systems by governments and the global community. This knowledge of the study area helps ensure that the proposed system is well aligned with the existing realities of the study area. Concerning data accessibility, the research utilizes a combination of public data sources, simulation data, as well as real-time sensor data. Public data sources pertaining to human detection, crowd counting, weather patterns, as well as disaster data, can be accessed from public research repositories as well as government sources. Furthermore, synthetic as well as simulated data will be generated with the objective of simulating different scenarios of disaster camps as well as different usages of disaster camp resources. Real-time data will be collected using IoT sensors as well as camera modules designed within a controlled experimental setup. Overall, the presence of relevant skill sets and the availability of domain-specific knowledge and data sources make the proposed study feasible. Integration of skill sets in smart technologies and the use of open data sources as well as the possible creation of real-time and simulated data provide a solid platform for the design and testing of the Smart IoT Box and its intelligent parts.

9. Objectives and Novelty

Main Objective Design and implement a Smart IoT Box for disaster relief camps that would integrate intelligent human and resource demand prediction, environmental risk monitoring, and offline education and awareness services using IoT, machine learning, and cloud technologies to improve the aspects of safety, efficiency, and decision-making in a limited connectivity environment.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Kashmira S S IT22059772	Development of an intelligent sub-component of a Smart IoT Box that can automatically estimate human population characteristics and predict essential resource demands in disaster relief camps by fusing IoT sensing with machine learning techniques.	Design an intelligent human and resource demand predicting module for a Smart IoT Box, which can enhance the efficacy and accuracy of resource allocation in relief camps during disasters and compare the resource demands predicted by the proposed technology with conventional manual resource estimation practices adopted by the disaster sector. • Find the appropriate IoT sensor devices (cameras, motion sensors) capable of handling low power and network consumption during the disaster scenario.	1. Integration of real-time human detection with demographic aware resource demand prediction, which is not prominently dealt with in the existing disaster management solutions. 2. Introduction of the concept of edge cloud hybrid intelligence to ensure functionality even with limited or low network connectivity. 3. Employment of the demographic composition of a population to enhance predictability of resource

		• Feasibility study for the application of Edge-based models for Real-time human detection and crowd counting using ML. • Develop a predictive algorithm that predicts demands on food, water, and medications in real time as well as in the future. • Integrate the human detection and resource prediction module with cloud-based dashboards.	demand rather than per person reasoning. 4. Application of predictive analytics to anticipate future shortages instead of reacting to existing deficits.
Deerasinghe A D S N S IT22065780	Low-Connectivity Adaptive Education and Disaster Awareness Intelligence: This component focuses on providing inclusive education support and disaster awareness to people living in disaster relief camps under low-network conditions. It provides not only	1. Requirement Analysis and Scope Definitions • Identify the educational needs of students in disaster relief camps. • Identify the awareness needs for students, adults, elderly individuals, and people with disabilities. • Define grade-wise and subject-wise learning requirements for students. 2. Component design • Design the architecture for a low-network learning application.	1. Introduces a unified approach that combines formal education continuity and disaster awareness within a single AI-driven system. 2. Focuses on inclusive and accessibility-aware content delivery, considering age diversity, disabilities, and literacy levels. 3. Unlike many AI education platforms that assume stable internet connectivity, this platform is designed

	personalized education, but also disaster-awareness content targeted at various age groups, levels of education, language, and disabilities. Using artificial intelligence, the component adapts to content complexity and presentation format (text, audio, and visual aids) to ensure information is easily understandable for all groups like children, adults, elderly individuals, and people with disabilities. The module integrates with the Smart-Camp IoT Box to function reliably in environments with	• Define interaction flow for students, elders, and camp administrators. • Collect basic user information to personalize content. ex: Age group, grade, subject and stream, preferred language, and accessibility needs (English/ Sinhala/ Tamil) 3. AI-based educational content generation design • Design prompts for grade-wise subject summarization. • Integrate large language model APIs for controlled educational content generation. • Ensure outputs are suitable for low bandwidth. 4. Disaster awareness content integration • Collect and structure disaster awareness and safety information. • Categorized content based on the disaster type. • Implement the logic to prioritize the awareness	specifically for low-network disaster environments. 4. Use human-centered AI adaptation instead of models that are performance-based or exam-oriented. 5. Contextually links educational content with real-world disaster situations to improve understanding and relevance for affected learners. 6. Extends the role of IoT in disaster camps beyond monitoring by supporting learning assistance and awareness.

	limited internet access while synchronizing with cloud services when connectivity is available.	content based on the disaster detected or predicted. 5. Machine Learning for personalization and optimization • Design lightweight machine learning techniques to adapt content complexity and the way of presentation based on user profiles and interaction patterns. • Use optimization logic and content recommendation to increase relevance for various age groups and educational levels. • Create demand prediction models to determine topics and subjects that are frequently accessed for effective local caching. 6. Low network optimization implementation • Implement local caching to enable continuous access to content during poor connectivity.	

		• Design fallback operational modes to ensure system functionality remains intact even in the event of a total network outage. • When intermittent connectivity becomes available, enable delayed and selective cloud synchronization. 7. Application development and IoT Integration • Develop an application interface that is easy to use and suitable for both non-technical users and students. • Integrate IoT edge devices used in disaster relief camps with the education and awareness component. • Ensure reliable communication between mobile applications, edge devices, and cloud services. 8. Evaluation and validation • Evaluate system performance. • Conduct scenario-based testing.	

		• Collect feedback and identify further system improvements.	
Hansana G P S IT22064868	Continuously monitors environmental conditions around the smart camp, triggers proactive risk mitigation actions, and prioritizes immediate alerts to survival teams upon detected or predicted disasters, even under low or unstable network connectivity.	1. Parameter Analysis and Scope Definition • Identify key environmental parameters relevant to smart camp safety. • Define acceptable threshold levels based on health, safety, and living conditions. • Scope monitoring areas such as classrooms, shelters, medical zones, and common spaces. 2. Device Selection & Component Design • Select suitable IoT sensors for each environmental parameter. • Plan sensor placement strategy and architecture to ensure full coverage and accuracy. • Define sensors' power, durability, and mobility	1. Integrates real-time environmental monitoring with AI-driven risk prediction specifically tailored to disaster relief camps. 2. Implements edge-first machine learning to detect hazards locally, ensuring continuous operation even during network outages. 3. Correlates multiple environmental parameters to provide compound risk assessment rather than single-factor alerts. 4. Pre-integrates all sensors within a self-contained disaster box, enabling autonomous operation without external intervention during emergencies.

		requirements suitable for camp conditions. • Design the architecture for a sensor and smart box connectivity. 3. Edge-Based Risk Detection Model Implementation • Find a suitable dataset with the required parameters. • Develop & train lightweight ML models to detect anomalies and unsafe environmental patterns. • Classify risk levels (normal, warning, critical) locally without cloud dependency. 4. Data Collection and Preprocessing Design • Identify and configure sensors for required environmental parameters. • Set up LAW-connected edge gateways to collect data continuously from all sensors. • Implement local preprocessing: filter noise,	5. Prioritizes environmental risk alerts from the Smart Disaster Box over other components upon network restoration, ensuring high-priority notifications reach authorities promptly, with hourly update cycles. 6. Synchronizes historical environmental data with cloud services selectively, enabling long-term analysis without requiring constant connectivity. 7. Supports proactive decision-making for camp administrators, linking environmental conditions directly to health, resource management, and learning components.

		normalize values, and handle missing data. • Verify sensor readings for consistency and detect faulty or outlier values. 5. Alerting and Response Mechanism Design • Define alert rules and priority levels for different risk scenarios. • Design local notification mechanisms for students, staff, and camp managers. • Specify recommended actions for each identified risk type. • Design alert escalation logic forwarded to higher authorities and cloud dashboards when connectivity is available. 6. Cloud Synchronization and Long-Term Analysis • Design delayed and selective data synchronization when connectivity is restored.	

		• Store historical environmental data for trend and pattern analysis. • Support centralized risk reporting and policy-level insights. 7. Environmental Data Visualization and Reporting • Design dashboards for camp to show environmental conditions and risk levels. • Implement graphical representation of sensor readings (charts, heatmaps, and trend lines). 8. Evaluation, Validation, and System Refinement • Test the component under simulated disaster and extreme environmental scenarios. • Evaluate accuracy, response time, and reliability of risk detection. • Collect stakeholder feedback and refine thresholds, models, and alerts.	

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Re organize individual component...

	Title	First Name	Last Name	Signature
Supervisor	Prof	Amarasiri	Dayaladda	
Co-Supervisor	Ms.	Sanchika	Lokuliyana	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				